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NEET · BIOLOGY · CHAPTER-WISE PYQS

Biotechnology: Principles and Processes

NEET Biology · Previous Year Questions with Answer Key

143	41	1994 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	142 questions
MCQ (More than One Correct Answer)	1 question
Answer key	at the end

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MCQ (Single Correct Answer) 142 questions

Q1. RE-NEET 2026

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : In recombinant DNA technology, lysozyme is used for disrupting bacterial cells while cellulase is for plant cells.

Reason R : Isolation of genetic material needs disruption of cells.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

(D) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct

Q2. RE-NEET 2026

Open solution

Match List-I with List-II.

	List-I		List-II
A.	Transformation	I.	Restriction enzyme
B.	Cloning site	II.	Transfer DNA to host bacteria
C.	Selection	III.	Replication
D.	Ori	IV.	Antibiotic

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II

(B) A-II, B-I, C-IV, D-III

(C) A-I, B-II, C-IV, D-III

(D) A-III, B-IV, C-II, D-I

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Q3. RE-NEET 2026

Open solution

Given below are two statements :

Statement I : Plasmids are autonomously replicating DNA.

Statement II: Plasmids are extrachromosomal DNA.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q4. RE-NEET 2026

Open solution

During PCR, primers bind to the DNA strands in the ____ step.

(A) ligation (B) denaturation (C) extension (D) annealing

Q5. NEET 2026

Open solution

Match List I with List II:

List-I		List-II	
A.	Genetically modified organism	I.	Agrobacterium tumefaciens
B.	Thermostable DNA polymerase	II.	Bt cotton
C.	Ti plasmid	III.	Thermus aquaticus
D.	pBR322	IV.	Escherichia coli

(A) A-II, B-III, C-I, D-IV (B) A-II, B-I, C-IV, D-III

(C) A-I, B-IV, C-III, D-II (D) A-I, B-II, C-IV, D-III

Q6. NEET 2026

Open solution

Identify the correct sequence of steps in each cycle of Polymerase Chain Reaction :

(A) Extension → Annealing → Denaturation (B) Annealing → Denaturation → Extension

(C) Denaturation → Extension → Annealing (D) Denaturation → Annealing → Extension

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Q7. NEET 2026

Open solution

Which of the following statements are not true regarding restriction endonucleases?

A. They are called molecular scissors.

B. These are the enzymes responsible for restricting the growth of bacteriophages in E. coli.

C. They cut the DNA only at the centre of the palindromic sites.

D. They remove nucleotides only from the ends of DNA fragments.

E. They recognise specific palindromic base-pair sequences.

Choose the answer from the options given below :

(A) A and B only **(B)** A and E only **(C)** D and E only **(D)** C and D only

Q8. NEET 2026

Open solution

Which of the following statements are correct with respect to DNA separation, isolation and visualization?

A. The cutting of DNA is done by molecular scissors.

B. The DNA fragments separate according to their size in an agarose gel, upon electrophoresis.

C. The separated DNA fragments can be seen without staining when exposed to UV light.

D. The separated DNA fragments, when stained with ethidium bromide, can be seen in visible light.

Choose the correct answer from the options given below :

(A) B and D only **(B)** A and B only **(C)** B and C only **(D)** A and D only

Q9. NEET 2026

Open solution

Insertion of a foreign DNA at BamHI site in an E.coli cloning vector pBR322 results in the loss of antibiotic resistance towards:

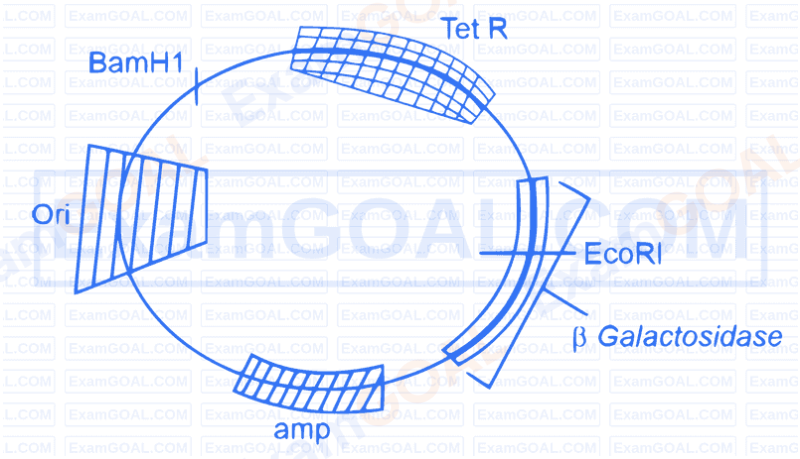
(A) Ampicillin and tetracycline **(B)** Ampicillin

(C) Tetracycline **(D)** Gentamycin

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Q10. NEET 2025

Open solution



In the above represented plasmid, an alien piece of DNA is inserted at EcoRI site. Which of the following strategies will be chosen to select the recombinant colonies?

(A) White color colonies will be selected.

(B) Blue color colonies grown on ampicillin plates can be selected.

(C) Using ampicillin & tetracycline containing medium plate.

(D) Blue color colonies will be selected.

Q11. NEET 2025

Open solution

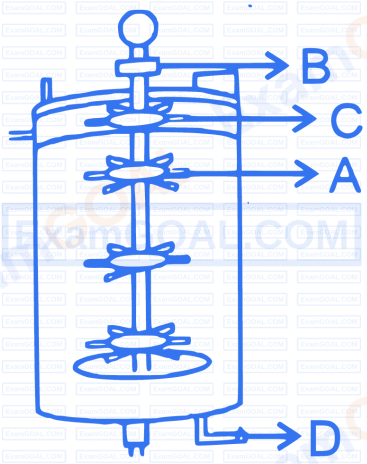
Polymerase chain reaction (PCR) amplifies DNA following the equation.

(A) $2n + 1$ (B) $2\,N^2$ (C) N^2 (D) 2^n

Q12. NEET 2025

Open solution

Identify the part of a bio-reactor which is used as a foam breaker from the given figure.



(A) D (B) C (C) A (D) B

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Q13. NEET 2025

The blue and white selectable markers have been developed which differentiate recombinant colonies from nonrecombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.

Open solution

Given below are two statements about this method:

Statement I : The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Statement II : The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Q14. NEET 2025

Which of the following enzyme(s) are NOT essential for gene cloning?

Open solution

A. Restriction enzymes

B. DNA ligase

C. DNA mutase

D. DNA recombinase

E. DNA polymerase

Choose the correct answer from the options given below:

(A) D and E only (B) B and C only (C) C and D only (D) A and B only

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Q15. NEET 2025

Given below are two statements:

Statement 1 : The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II : Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but statement II is correct

(C) Both statement I and statement II are correct

(D) Both statement I and statement II are incorrect

Open solution

Q16. NEET 2024 (Re-Examination)

Identify the incorrect statement related to gel electrophoresis.

(A) Separated DNA fragments can be directly seen under UV radiation

(B) Separated DNA can be extracted from gel piece

(C) Fragment of DNA moves toward anode

(D) Sieving effect of agarose gel helps in separation of DNA fragments

Open solution

Q17. NEET 2024 (Re-Examination)

Recombinant DNA molecule can be created normally by cutting the vector DNA and source DNA respectively with:

(A) Hind II, Hind II (B) Hind II, Alu I (C) Hind II, EcoR I (D) Hind II, BamHI

Open solution

Q18. NEET 2024 (Re-Examination)

Select the restriction endonuclease enzymes whose restriction sites are present for the tetracycline resistance (tet^R) gene in the pBR322 cloning vector.

(A) Bam HI and Sal I (B) Sal I and Pst I (C) Pst I and Pvu I (D) Pvu I and Bam HI

Open solution

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Q19. NEET 2024 (Re-Examination)

Open solution

Which of the following are correct about EcoRI?

A. Cut the DNA with blunt end

B. Cut the DNA with sticky end

C. Recognise a specific palindromic sequence

D. Cut the DNA between the base G and A when encounters the DNA sequence 'GAATTC'

E. Exonuclease

Choose the correct answer from the options given below:

(A) B, C, E only **(B)** A, D, E only **(C)** A, C, D only **(D)** B, C, D only

Q20. NEET 2024 (Re-Examination)

Open solution

Following are the steps involved in the process of PCR.

A. Annealing

B. Amplification (~1 billion times)

C. Denaturation

D. Treatment with Taq polymerase and deoxynucleotides

E. Extension

Choose the correct sequence of steps of PCR from the options given below :

(A) $C \rightarrow A \rightarrow D \rightarrow E \rightarrow B$

(B) $A \rightarrow B \rightarrow E \rightarrow D \rightarrow C$

(C) $A \rightarrow C \rightarrow E \rightarrow D \rightarrow B$

(D) $D \rightarrow B \rightarrow E \rightarrow C \rightarrow A$

Q21. NEET 2024

Open solution

Hind II always cuts DNA molecules at a particular point called recognition sequence and it consists of:

(A) 8 bp **(B)** 6 bp **(C)** 4 bp **(D)** 10 bp

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Q22. NEET 2024

What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?

Open solution

A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.

B. It may get integrated into the genome of the recipient.

C. It may multiply and be inherited along with the host DNA.

D. The alien piece of DNA is not an integral part of chromosome.

E. It shows ability to replicate.

Choose the correct answer from the options given below:

(A) A and B only (B) D and E only (C) B and C only (D) A and E only

Q23. NEET 2024

The "Ti plasmid" of Agrobacterium tumefaciens stands for

Open solution

(A) Tumour inhibiting plasmid (B) Tumor independent plasmid

(C) Tumor inducing plasmid (D) Temperature independent plasmid

Q24. NEET 2024

Which of the following statements is incorrect?

Open solution

(A) A bio-reactor provides optimal growth conditions for achieving the desired product

(B) Most commonly used bio-reactors are of stirring type

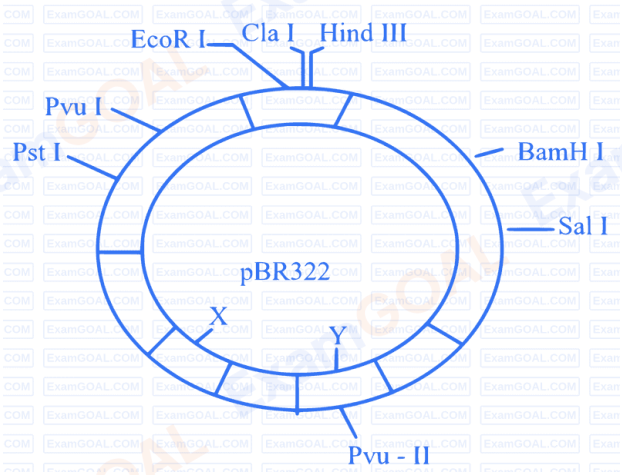
(C) Bio-reactors are used to produce small scale bacterial cultures

(D) Bio-reactors have an agitator system, an oxygen delivery system and foam control system

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Q25. NEET 2024

The following diagram shows restriction sites in E. coli cloning vector pBR322. Find the role of 'X' and 'Y' genes :



Open solution

(A) The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.

(B) The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.

(C) The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.

(D) Gene 'X' is responsible for recognition sites and 'Y' is responsible for antibiotic resistance.

Q26. NEET 2023 Manipur

Thermostable DNA polymerase used in PCR was isolated from :

(A) Thermus aquaticus

(B) Escherichia coli

(C) Agrobacterium tumifaciens

(D) Bacillus thuringiensis

Open solution

Q27. NEET 2023 Manipur

Ligation of foreign DNA at which of the following site will result in loss of tetracyclin resistance of pBR322 :

(A) Pst I

(B) Pvu I

(C) EcoR I

(D) BamH I

Open solution

Q28. NEET 2023 Manipur

Which of the following statement is incorrect about Agrobacterium tumifaciens?

(A) It is used to deliver gene of interest in both prokaryotic as well as eukaryotic host cells.

(B) 'Ti' plasmid from Agrobacterium tumifaciens used for gene transfer is not pathogenic to plant cell.

(C) It transforms normal plant cells into tumor cells.

(D) It delivers 'T-DNA' into plant cell.

Open solution

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Q29. NEET 2023 Manipur

Match List I with List II.

	List - I		List - II
(A)	Kanamycin	(I)	Delivers genes into animal cells
(B)	ClaI	(II)	Selectable marker
(C)	Disarmed retroviruses	(III)	Restriction site
(D)	Kanamycin ^{R}gene	(IV)	Antibiotic resistance

Choose the **correct** answer from the options given below :

(A) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
(B) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
(D) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Open solution

Q30. NEET 2023 Manipur

Which of the following can act as molecular scissors?

(A) Restriction enzymes
(B) DNA ligase
(C) RNA polymerase
(D) DNA polymerase

Open solution

Q31. NEET 2023

During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out

(A) DNA **(B)** Histones **(C)** Polysaccharides **(D)** RNA

Open solution

Q32. NEET 2023

In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.

(A) Zinc **(B)** Tungsten or gold **(C)** Silver **(D)** Copper

Open solution

Q33. NEET 2023

Upon exposure to UV radiation, DNA stained with ethidium bromide will show

(A) Bright blue colour
(B) Bright yellow colour
(C) Bright orange colour
(D) Bright red colour

Open solution

Q34. NEET 2023

Which of the following is not a cloning vector?

(A) YAC **(B)** pBR322 **(C)** Probe **(D)** BAC

Open solution

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Q35. NEET 2022 Phase 2

Open solution

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : When a particular restriction enzyme cuts strand of DNA, overhanging stretches or sticky ends are formed.

Reason (R) : Some restriction enzymes cut the strand of DNA a little away from the centre of palindromic site.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is not correct but (R) is correct

(B) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) (A) is correct but (R) is not correct

Q36. NEET 2022 Phase 2

Open solution

Separation of DNA fragments is done by a technique known as

(A) Gel electrophoresis

(B) Polymerase Chain Reaction

(C) Recombinant technology

(D) Southern blotting

Q37. NEET 2022 Phase 2

Open solution

The enzyme (a) is needed for isolating genetic material from plant cells and enzyme (b) for isolating genetic material from fungus. Choose the correct pair of options from the following :

(A) (a) Cellulase (b) Lipase

(B) (a) Cellulase (b) Protease

(C) (a) Cellulase (b) Chitinase

(D) (a) Chitinase (b) Lipase

Q38. NEET 2022 Phase 2

Open solution

Match List-I with List-II :

	List - I		List - II
(a)	Gene gun	(i)	Replacement of a faulty gene by a normal healthy gene
(b)	Gene therapy	(ii)	Used for transfer of gene
(c)	Gene cloning	(iii)	Total DNA in the cells of an organism
(d)	Genome	(iv)	To obtain identical copies of a particular DNA molecule

Choose the correct answer from the options given below:

(A) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(B) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

(C) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

(D) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

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Q39. NEET 2022 Phase 2

Pathogenic bacteria gain resistance to antibiotics due to changes in their :

(A) Nucleoid (B) Cosmids (C) Plasmids (D) Nucleus

Open solution

Q40. NEET 2022 Phase 2

Which of the following methods is not commonly used for introducing foreign DNA into the plant cell?

(A) Bacteriophages
(B) Agrobacterium mediated transformation
(C) Gene gun
(D) 'Disarmed pathogen' vectors

Open solution

Q41. NEET 2022 Phase 2

Refer to the following statements for agarose-gel electrophoresis:

(a) Agarose is a natural polymer obtained from sea-weed.

(b) The separation of DNA molecules in agarose-gel electrophoresis depends on the size of DNA.

(c) The DNA migrates from negatively-charged electrode to the positively-charged electrode

(d) The DNA migrates from positively-charged electrode to the negatively-charged electrode.

Choose the most appropriate answer from the options given below :

(A) (b), (c) and (d) only
(B) (a) and (b) only
(C) (a), (b) and (c) only
(D) (a), (b) and (d) only

Open solution

Q42. NEET 2022 Phase 1

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Polymerase chain reaction is used in DNA amplification.

Reason (R) : The ampicillin resistant gene is used as a selectable marker to check transformation.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(C) (A) is correct but (R) is not correct
(D) (A) is not correct but (R) is correct

Open solution

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Q43. NEET 2022 Phase 1

Which one of the following statement is not true regarding gel electrophoresis technique?

(A) The process of extraction of separated DNA strands from gel is called elution.

(B) The separated DNA fragments are stained by using ethidium bromide.

(C) The presence of chromogenic substrate gives blue coloured DNA bands on the gel.

(D) Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.

Open solution

Q44. NEET 2022 Phase 1

In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme?

(A) 5'GATACT3' ; 3'CTATGA5'

(B) 5'GAATTC3' ; 3'CTTAAG5'

(C) 5'CTCAGT3' ; 3'GAGTCA5'

(D) 5'GTATTC3' ; 3'CATAAG5'

Open solution

Q45. NEET 2022 Phase 1

Given below are two statements:

Statement I : Restriction endonucleases recognise specific sequence to cut DNA known as palindromic nucleotide sequence.

Statement II : Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Open solution

Q46. NEET 2022 Phase 1

Which of the following is not a desirable feature of a cloning vector?

(A) Presence of origin of replication

(B) Presence of a marker gene

(C) Presence of single restriction enzyme site

(D) Presence of two or more recognition sites

Open solution

Q47. NEET 2021

During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out :

(A) Polysaccharides (B) RNA (C) DNA (D) Histones

Open solution

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Q48. NEET 2021

DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as

(A) Bright blue bands
(B) Yellow bands
(C) Bright orange bands
(D) Dark red bands

Open solution

Q49. NEET 2021

Which of the following is not an application of PCR (Polymerase Chain Reaction)?

(A) Detection of gene mutation
(B) Molecular diagnosis
(C) Gene amplification
(D) Purification of isolated protein

Open solution

Q50. NEET 2021

Which of the following is a correct sequence of steps in a PCR (Polymerase Chain Reaction)?

(A) Annealing, Denaturation, Extension
(B) Denaturation, Annealing, Extension
(C) Denaturation, Extension, Annealing
(D) Extension, Denaturation, Annealing

Open solution

Q51. NEET 2021

Plasmid pBR322 has Pst I restriction enzyme site within gene amp^R that confers ampicillin resistance. If this enzyme is used for inserting a gene for β-galactoside production and the recombinant plasmid is inserted in an E.coli strain

(A) It will be able to produce a novel protein with dual ability
(B) It will not be able to confer ampicillin resistance to the host cell
(C) The transformed cells will have the ability to resist ampicillin as well as produce β-galactoside
(D) It will lead to lysis of host cell

Open solution

Q52. NEET 2021

During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first?

(A) Ligation
(B) Annealing
(C) Extension
(D) Denaturation

Open solution

Q53. NEET 2021

A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :

(A) Poly (A) tail sequences
(B) Degenerate primer sequence
(C) Okazaki sequences
(D) Palindromic Nucleotide sequences

Open solution

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Q54. NEET 2020 Phase 2

Match the following techniques or instruments with their usage :

(a) Bioreactor	(i) Separation of DNA fragments
(b) Electrophoresis	(ii) Production of large quantities of products
(c) PCR	(iii) Detection of pathogen, based on antigen - antibody reaction
(d) ELISA	(iv) Amplification of nucleic acids

Select the correct option from following :

(A) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
(B) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
(C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(D) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Open solution

Q55. NEET 2020 Phase 2

In a mixture, DNA fragments are separated by

(A) Polymerase chain reaction
(B) Bioprocess engineering
(C) Restriction digestion
(D) Electrophoresis

Open solution

Q56. NEET 2020 Phase 2

In Recombinant DNA technology antibiotics are used :

(A) as selectable markers
(B) to keep medium bacteria-free
(C) to detect alien DNA
(D) to impart disease-resistance to the host plant

Open solution

Q57. NEET 2020 Phase 2

First discovered restriction endonuclease that always cuts DNA molecule at a particular point by recognising a specific sequence of six base pairs is :

(A) Hind II
(B) EcoRI
(C) Adenosine deaminase
(D) Thermostable DNA polymerase

Open solution

Q58. NEET 2020 Phase 2

Select the correct statement from the following :

(A) PCR is used for isolation and separation of gene of interest.
(B) Gel electrophoresis is used for amplification of a DNA segment.
(C) The polymerase enzyme joins the gene of interest and the vector DNA.
(D) Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzymes of optimum conditions.

Open solution

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Q59. NEET 2020 Phase 2

Spooling is -

(A) Collection of isolated DNA

(B) Amplification of DNA

(C) Cutting of separated DNA bands from the agarose gel

(D) Transfer of separated DNA fragments to synthetic membranes

Open solution

Q60. NEET 2020 Phase 1

In gel electrophoresis, separated DNA fragments can be visualized with the help of

(A) Ethidium bromide in infrared radiation

(B) Ethidium bromide in UV radiation

(C) Acetocarmine in bright blue light

(D) Acetocarmine in UV radiation

Open solution

Q61. NEET 2020 Phase 1

Identify the wrong statement with regard to Restriction Enzymes

(A) Sticky ends can be joined by using DNA ligases.

(B) They cut the strand of DNAat palindromic sites.

(C) Each restriction enzyme functions by inspecting the length of a DNA sequence.

(D) They are useful in genetic engineering

Open solution

Q62. NEET 2020 Phase 1

The specific palindromic sequence which is recognized by EcoRI is

(A) 5' - GGATCC - 3'
3' - CCTAGG - 5'

(B) 5' - GAATTC - 3'
3' - CTTAAG - 5'

(C) 5' - GGAACC - 3'
3' - CCTTGG - 5'

(D) 5' - CTTAAG - 3'
3' - GAATTC - 5'

Open solution

Q63. NEET 2020 Phase 1

The sequence that controls the copy number of the linked DNA in the vector, is termed

(A) Recognition site

(B) Ori site

(C) Selectable marker

(D) Palindromic sequence

Open solution

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Q64. NEET 2020 Phase 1

Match the organism with its use in biotechnology

Open solution

(a) Bacillus thuringiensis	(i) Cloning vector
(b) Thermus aquaticus	(ii) Construction of first rDNA molecule
(c) Agrobacterium tumefaciens	(iii) DNA polymerase
(d) Salmonella typhimurium	(iv) Cry proteins

Select the correct option from the following:

(A) (A) → (iii), (B) → (iv), (C) → (i), (D) → (ii)

(B) (A) → (ii), (B) → (iv), (C) → (iii), (D) → (i)

(C) (A) → (iv), (B) → (iii), (C) → (i), (D) → (ii)

(D) (A) → (iii), (B) → (ii), (C) → (iv), (D) → (i)

Q65. NEET 2020 Phase 1

Choose the correct pair from the following

Open solution

(A) Exonucleases – Make cuts at specific positions within DNA

(B) Ligases – Join the two DNA molecules

(C) Polymerases – Break the DNA into fragments

(D) Nucleases – Separate the two strands of DNA

Q66. NEET 2019 (Odisha)

A selectable marker is used to :

Open solution

(A) Mark a gene on a chromosome for isolation using restriction enzyme

(B) Help in eliminating the non-transformants, so that the transformants can be regenerated

(C) Identify the gene for a desired trait in an alien organism

(D) Select a suitable vector for transformation in a specific crop

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Q67. NEET 2019 (Odisha)

Given below are four statements pertaining to separation of DNA fragments using Gel electrophoresis. Identify the incorrect statements.

(a) DNA is negatively charged molecule and so it is loaded on gel towards the Anode terminal.

(b) DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA.

(c) Smaller the size of DNA fragment, larger is the distance it travels through it.

(d) Pure DNA can be visualized directly by exposing to UV radiation.

Choose correct answer from the options given below :

(A) (a), (b) and (d) **(B)** (a), (c) and (d) **(C)** (a), (b) and (c) **(D)** (b), (c) and (d)

Open solution

Q68. NEET 2019 (Odisha)

An enzyme catalysing the removal of nucleotides from ends of DNA is :

(A) Protease **(B)** DNA ligase **(C)** Endonuclease **(D)** Exonuclease

Open solution

Q69. NEET 2019 (Odisha)

Match the following enzymes with their functions:

a. Restriction endonuclease	(i) Joins the DNA fragments
b. Restriction exonuclease	(ii) Extends primers on genomic DNA template
c. DNA ligase	(iii) Cuts DNA at specific position
d. Taq polymerase	(iv) Removes nucleotides from the ends of DNA

Select the correct option from the following:

(A) a(ii), b(iv), c(i), d(iii) **(B)** a(iii), b(i), c(iv), d(ii)

(C) a(iii), b(iv), c(i), d(ii) **(D)** a(iv), b(iii), c(i), d(ii)

Open solution

Q70. NEET 2019 (Odisha)

The two antibiotic resistance genes on vector pBR322 are for

(A) Tetracycline and Kanamycin **(B)** Ampicillin and Tetracycline

(C) Ampicillin and Chloramphenicol **(D)** Chloramphenicol and Tetracycline

Open solution

Q71. NEET 2019

DNA precipitation out of a mixture of biomolecules can be achieved by treatment with :

(A) Methanol at room temperature **(B)** Chilled chloroform

(C) Isopropanol **(D)** Chilled ethanol

Open solution

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Q72. NEET 2019

Following statements describe the characteristics of the enzyme Restriction Endonuclease. Identify the **incorrect** statement.

(A) The enzyme cuts the sugar-phosphate backbone at specific sites on each strand

(B) The enzyme recognizes a specific palindromic nucleotide sequence in the DNA

(C) The enzyme cuts DNA molecule at identified position within the DNA

(D) The enzyme binds DNA at specific sites and cuts only one of the two strands

Open solution

Q73. NEET 2018

The correct order of steps in Polymerase Chain Reaction (PCR) is

(A) extension, denaturation, annealing

(B) annealing, extension, denaturation

(C) denaturation, extension, annealing

(D) denaturation, annealing, extension

Open solution

Q74. NEET 2017

The DNA fragments separated on an agarose gel can be visualised after staining with

(A) Bromophenol blue

(B) Aniline blue

(C) Ethidium bromide

(D) Acetocarmine

Open solution

Q75. NEET 2017

DNA fragments are

(A) Positively charged

(B) Either positively or negatively charged depending on their size

(C) Neutral

(D) Negatively charged

Open solution

Q76. NEET 2017

A gene whose expression helps to identify transformed cell is known as

(A) Selectable marker

(B) Plasmid

(C) Vector

(D) Structural gene

Open solution

Q77. NEET 2017

What is the criterion for DNA fragments movement on agarose gel during gel electrophoresis ?

(A) Positively charged fragment move to farther end

(B) Negatively charged fragments do not move

(C) The smaller the fragment size, the farther it moves

(D) The larger the fragment size, the farther it moves

Open solution

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Q78. NEET 2017

The process of separation and purification of expressed protein before marketing is called :

(A) Upstream processing
(B) Bioprocessing
(C) Downstream processing
(D) Postproduction processing

Open solution

Q79. NEET 2016 Phase 2

Stirred-tank bioreactors have been designed for -

(A) ensuring anaerobic conditions in the culture vessel
(B) addition of preservatives to the product
(C) purification of product
(D) availability of oxygen throughout the process

Open solution

Q80. NEET 2016 Phase 2

A foreign DNA and plasmid cut by the same restriction endonuclease can be joined to form a recombinant plasmid using

(A) Eco RI
(B) Polymerase III
(C) Ligase
(D) Taq polymerase

Open solution

Q81. NEET 2016 Phase 2

Which of the following is not a component of downstream processing ?

(A) Purification
(B) Expression
(C) Preservation
(D) Separation

Open solution

Q82. NEET 2016 Phase 2

Which of the following restriction enzymes produces blunt ends ?

(A) Eco RV
(B) Hind III
(C) Xho I
(D) Sal I

Open solution

Q83. NEET 2016 Phase 1

Which of the following is not a feature of the plasmids ?

(A) Single-stranded
(B) Transferable
(C) Independent replication
(D) Circular structure

Open solution

Q84. NEET 2016 Phase 1

The taq polymerase enzyme is obtained from

(A) Thermus aquaticus
(B) Bacillus subtilis
(C) Pseudomonas putida
(D) Thiobacillus ferrooxidans

Open solution

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Q85. NEET 2016 Phase 1

Which of the following is a restriction endonuclease ?

(A) DNase I (B) Hind II (C) RNase I (D) Protease

Open solution

Q86. AIPMT 2015

The DNA molecule to which the gene of interest is integrated for cloning is called -

(A) Transformer (B) Template (C) Vector (D) Carrier

Open solution

Q87. AIPMT 2015

The cutting of DNA at specific locations became possible with the discovery of -

(A) Restriction enzymes (B) Selectable markers

(C) Probes (D) Ligases

Open solution

Q88. AIPMT 2014

An analysis of chromosomal DNA using the Southern hybridization technique does not use:

(A) Blotting (B) PCR (C) Electrophoresis (D) Autoradiography

Open solution

Q89. AIPMT 2014

Which vector can clone only a small fragment of DNA ?

(A) Cosmid (B) Bacterial artificial chromosome

(C) Yeast artificial chromosome (D) Plasmid

Open solution

Q90. AIPMT 2014

Commonly used vectors for human genome sequencing are :

(A) BAC and YAC (B) T/A Cloning Vectors

(C) T− DNA (D) Expression Vectors

Open solution

Q91. NEET 2013 (Karnataka)

During the process of isolation of DNA, chilled ethanol is added to :

(A) Break open the cell to release DNA (B) Remove proteins such as histones

(C) Facilitate action of restriction enzymes (D) Precipitate DNA

Open solution

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Q92. NEET 2013

The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of :

(A) Insertional inactivation of alphagalactosidase in recombinant bacteria

(B) Inactivation of glycosidase enzyme in recombinant bacteria

(C) Insertional inactivation of alphagalactosidase in non-recombinant bacteria

(D) Non-recombinant bacteria containing betagalactosidase

Open solution

Q93. NEET 2013

DNA fragments generated by the restriction endonucleases in a chemical reaction can be separated by :

(A) Polymerase chain reaction

(B) Restriction mapping

(C) Electrophoresis

(D) Centrifugation

Open solution

Q94. NEET 2013

Which of the following is not corretly matched for the organism and its cell wall degrading enzyme ?

(A) Plant cells- Cellulase

(B) Algae-Methylase

(C) Fungi - Chitinase

(D) Bacteria-Lysozyme

Open solution

Q95. AIPMT 2012 Mains

Biolistics (gene-gun) is suitable for -

(A) DNA finger printing

(B) Constructing recombinant DNA by joining with vectors

(C) Disarming pathogen vectors

(D) Transformation of plant cells

Open solution

Q96. AIPMT 2012 Mains

In genetic engineering, the antibiotics are used

(A) As sequences from where replication starts

(B) As selectable markers

(C) To keep the cultures free of infection

(D) To select healthy vectors

Open solution

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Q97. AIPMT 2012 Mains

Which one of the following represents a palindromic sequence in DNA ?

(A) 5'-GAATTC-3'
3'-CTTAAG-5'

(B) 5'-GATACC-3'
3'-CCTAAG-5'

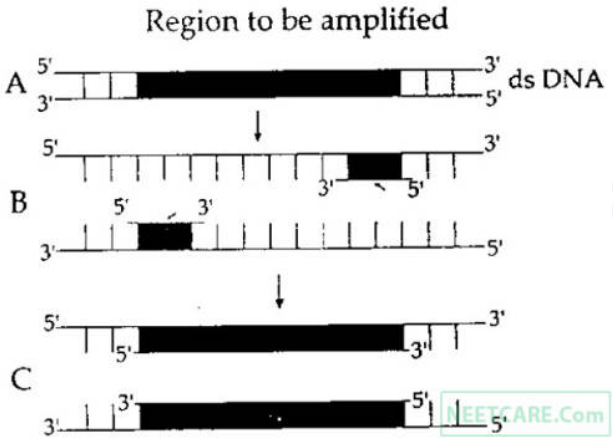
(C) 5'-CCAATG-3'
3'-GAATCC-5'

(D) 5'-CATTAG-3'
3'-GATAAC-5'

Open solution

Q98. AIPMT 2012 Mains

The figure below shows three steps (A, B, C) of Polymerase Chain Reaction (PCR). Select the option giving correct identification together with what it represents ?



Open solution

(A) C - Extension in the presence of heat stable DNA polymerase

(B) B - Denaturation at a temperature of about 98°C separating the two DNA strands

(C) A - Denaturation at a temperature of about 50°C

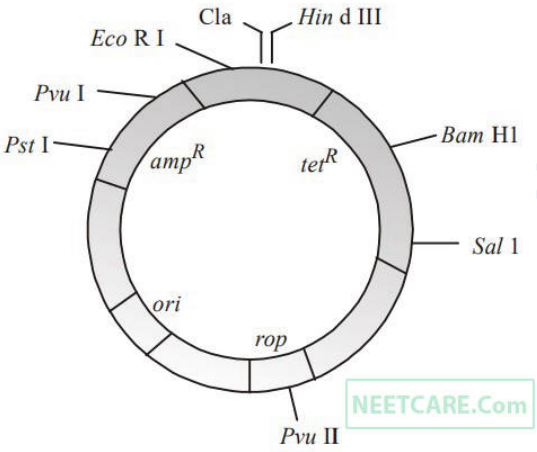
(D) A - Annealing with two sets of primers

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Q99. AIPMT 2012 Prelims

The figure below is the diagrammatic representation of the E. coli vector pBR 322. Which one of the given options correctly identifies its certain component(s) ?

Open solution



(A) ori- original restriction enzyme

(B) amp^R, tet^R-antibiotic resistance genes

(C) rop-reduced osmotic pressure

(D) Hind III, EcoRI-selectable markers

Q100. AIPMT 2012 Prelims

PCR and Restriction Fragment Length Polymorphism are the methods for

Open solution

(A) Genetic transformation

(B) Genetic Fingerprinting

(C) Study of enzymes

(D) DNA sequencing

Q101. AIPMT 2012 Prelims

Which one is a true statement regarding DNA polymerase used in PCR ?

Open solution

(A) It is used to ligate introduced DNA in recipient cell

(B) It remains active at high temperature

(C) It serves as a selectable marker

(D) It is isolated from a virus

Q102. AIPMT 2012 Prelims

For transformation, micro - particles coated with DNA to be bombarded with gene are made up of

Open solution

(A) Gold or Tungsten

(B) Platinum or zinc

(C) Silver or platinum

(D) Silicon or platinum

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Q103. AIPMT 2011 Mains

Which one of the following techniques made it possible to genetically engineer living organisms ?

(A) Recombinant DNA techniques
(B) X-ray diffraction
(C) Heavier isotope labeling
(D) Hybridization

Open solution

Q104. AIPMT 2011 Prelims

Continuous addition of sugars in 'fed batch' fermentation is done to :

(A) obtain antibiotics
(B) produce methane
(C) purify enzymes
(D) degrade sewage

Open solution

Q105. AIPMT 2011 Prelims

Given below is a sample of a portion of DNA strand giving the base sequence on the opposite strands. What is so special shown in it ?
5' ..... GAATTC ..... 3'
3' ..... CTTAAG ..... 5'

(A) Start codon at the 5' end
(B) Replication completed
(C) Deletion mutation
(D) Palindromic sequence of base pairs

Open solution

Q106. AIPMT 2011 Prelims

There is a restriction endonuclease called EcoRI. What does "co" part in it stand for ?

(A) colon
(B) coelom
(C) coenzyme
(D) coli

Open solution

Q107. AIPMT 2011 Prelims

Agarose extracted from sea weeds finds use in :

(A) Spectrophotometry
(B) PCR
(C) Tissue Culture
(D) Gel electrophoresis

Open solution

Q108. AIPMT 2010 Mains

Which of the following are used in gene cloning ?

(A) Mesosomes
(B) Nucleoids
(C) Lomasomes
(D) Plasmids

Open solution

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Q109. AIPMT 2010 Mains

In genetic engineering, a DNA segment (gene) of interest, is transferred to the host cell through a vector. Consider the following four agents (A-D) in this regard and select the correct option about which one or more of these can be used as a vector / vectors –

(A) a bacterium
(B) plasmid
(C) plasmodium
(D) bacteriophage

(A) (B) and (D) only
(B) (A) and (C) only
(C) (A), (B) and (D) only
(D) (A) only

Open solution

Q110. AIPMT 2010 Prelims

Which one of the following palindromic base sequences in DNA can be easily cut at about the middle by some particular restriction enzyme?

(A) 5' _____ CG TTCG _____ 3'
3' _____ATGGTA _____5'

(B) 5'_____GAATTC _____3'
3'_____CTTAAG_____5'

(C) 5'_____CACGTA_____3'
3'_____CTCAGT_____5'

(D) 5'_____GATATG _____3'
3'_____CTACTA _____5'

Open solution

Q111. AIPMT 2010 Prelims

Which one of the following is used as vector for cloning genes into higher organisms ?

(A) Retrovirus
(B) Salmonella typhimurium
(C) Rhizopus nigricans
(D) Baculovirus

Open solution

Q112. AIPMT 2010 Prelims

Restriction endonucleases are enzymes which –

(A) recognize a specific nucleotide sequence for binding of DNA ligase
(B) remove nucleotides from the ends of the DNA molecule
(C) restrict the action of the enzyme DNA polymerase
(D) make cuts at specific positions within the DNA molecule

Open solution

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Q113. AIPMT 2010 Prelims

Stirred-tank bioreactors have been designed for –

(A) Purification of the product

(B) Availability of oxygen throughout the process

(C) Ensuring anaerobic conditions in the culture vessel

(D) Addition of preservatives to the product

Open solution

Q114. AIPMT 2009

Which one of the following is commonly used in transfer of foreign DNA into crop plants ?

(A) Trichoderma harzianum

(B) Meloidogyne incognita

(C) Agrobacterium tumefaciens

(D) Penicillium expansum

Open solution

Q115. AIPMT 2008

Gel electrophoresis is used for:

(A) cutting of DNA into fragments

(B) separation of DNA fragments according to their size

(C) construction of recombinant DNA by joining with cloning vectors

(D) isolation of DNA molecule

Open solution

Q116. AIPMT 2008

The linking of antibiotic resistance gene with the plasmid vector became possible with:

(A) Exonucleases (B) Endonucleases (C) DNA ligase (D) DNA polymerase

Open solution

Q117. AIPMT 2006

Restriction endonuclease -

(A) Cuts the DNA molecule at specific sites

(B) Synthesizes DNA

(C) Restricts the synthesis of DNA inside the nucleus

(D) Cuts the DNA molecule randomly

Open solution

Q118. AIPMT 2006

Two microbes found to be very useful in genetic engineering are-

(A) Diplococcus sp.and Pseudomonas sp.

(B) Vibrio cholerae and a tailed bacteriophage

(C) Crown gall bacterium and Caenorhabditis elegans

(D) Escherichia coli and Agrobacterium tumefaciens

Open solution

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Q119. AIPMT 2005

Which one of the following hydrolyses internal phosphodiester bonds in a polynucleotide chain ?

(A) protease (B) Endonuclease (C) Lipase (D) Exonuclease

Open solution

Q120. AIPMT 2004

Restriction endonucleases :-

(A) Are used for in vitro DNA synthesis

(B) Are present in mammalian cells for degradation of DNA when the cell dies

(C) Are synthesized by bacteria as part of their defense mechanism

(D) Are used in genetic engineering for ligating two DNA molecules

Open solution

Q121. AIPMT 2004

In transgenics expression of transgene in target tissue is determined by :-

(A) Enhancer (B) Promoter (C) Reporter (D) Transgene

Open solution

Q122. AIPMT 2004

The Ti plasmid is often used for making transgenic plants. This plasmid is found in

(A) Agrobacterium

(B) Rhizobium of the roots of leguminous plants

(C) Azotobacter

(D) Yeast as a 2 μm plasmid

Open solution

Q123. AIPMT 2003

In recent years, DNA sequences (nucleotide sequence) of mt-DNA and Y chromosomes were considered for the study of human evolution, because : -

(A) They are uniparental in origin and do not take part in recombination

(B) They can be studied from the samples of fossil remains

(C) Their structure is known in greater detail

(D) They are small, and therefore, easy to study

Open solution

Q124. AIPMT 2003

Which one of the following bacteria has found extensive use in genetic engineering work in plants ?

(A) Xanthomonas citri (B) Agrobacterium tumefaciens

(C) Bacillus coagulens (D) Clostridium septicum

Open solution

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Q125. AIPMT 2002

Manipulation of DNA in genetic engineering became possible due to the discovery of : -

(A) Transcriptase
(B) DNA ligase
(C) Primase
(D) Restriction endonuclease

Open solution

Q126. AIPMT 2002

Which steroid is used for transformation?

(A) Cortisol
(B) Cholesterol
(C) Testosterone
(D) Progesterone

Open solution

Q127. AIPMT 2001

A mutant strain of T4 – Bacteriophage, R-II, fails to lyse the E-Coli but when two strains RIIX and R-IIY are mixed then they lyse the E.Coli. What may be the possible reason : -

(A) Both strains have different cistrons
(B) Both strains have simillar cistrons
(C) It is not mutated
(D) Bacteriophage transforms in wild

Open solution

Q128. AIPMT 2001

In lederberg's replica plating experiment what shall be used to obtain streptomycin resistant strain : -

(A) Only minimal medium
(B) Complete medium and streptomycine
(C) Only complete medium
(D) Minimal medium and streptomycine

Open solution

Q129. AIPMT 2001

Which of the following cut the DNA from specific places ?

(A) Exonuclease
(B) Alkaline phosphate
(C) E.coli restriction endonuclease I
(D) Ligase

Open solution

Q130. AIPMT 2001

Maximum number of bases in plasmids discovered so far : -

(A) 500 kilo base
(B) 5 kilo base
(C) 50 kilo base
(D) 5000 kilo base

Open solution

Q131. AIPMT 2001

Plasmid is

(A) fragment of DNA which acts as vector
(B) a fragment which joins two genes
(C) m-RNA which acts as carrier
(D) autotrophic fragment

Open solution

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Q132. AIPMT 2000

The bacteria generally used for genetic engineering is :

(A) Clostridium (B) Pseudomonas (C) Agrobacterium (D) Bacillus

Open solution

Q133. AIPMT 2000

Plasmid has been used as vector because :

(A) Both ends show replication

(B) It has antibiotic resistance gene

(C) It is circular DNA which have capacity to join to eukaryotic DNA

(D) It can move between prokaryotic and eukaryotic cells

Open solution

Q134. AIPMT 1999

The process of replication in plasmid DNA, other than initiation, is controlled by

(A) mitochondrial gene (B) bacterial gene

(C) plasmid gene (D) none of the above

Open solution

Q135. AIPMT 1999

Which of the following is related to genetic engineering?

(A) Mutation (B) Plasmid (C) Plastid (D) Heterosis

Open solution

Q136. AIPMT 1998

Recombinant DNA is obtained by cleaving the pro-DNAs by

(A) primase (B) exonucleases

(C) ligase (D) restriction endonuclease

Open solution

Q137. AIPMT 1998

The chemical knives of DNA are

(A) ligases (B) polymerases (C) endonucleases (D) transcriptases

Open solution

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Q138. AIPMT 1998

Restriction endonucleases are

(A) synthesized by dog

(B) present in mammalian cells for degradation of DNA

(C) used for *in vitro* DNA synthesis

(D) used in genetic engineering

Open solution

Q139. AIPMT 1998

Genetic engineering is possible, because

(A) the phenomenon of transduction in bacteria is well understood

(B) we can see DNA by electron microscope

(C) we can cut DNA at specific sites by endonucleases like DNAse I

(D) restriction endonucleases purified from bacteria can be used *in vitro*

Open solution

Q140. AIPMT 1998

Two bacteria found to be very useful in genetic engineering experiments are

(A) *Nitrosomonas* and *Klebsiella*

(B) *Escherichia* and *Agrobacterium*

(C) *Nitrobacter* and *Azotobacter*

(D) *Rhizobium* and *Diplococcus*

Open solution

Q141. AIPMT 1995

The restriction enzymes are used in genetic engineering, because

(A) they can cut DNA at specific base sequence

(B) they are nucleases that cut DNA at variable sites

(C) they can degrade harmful proteins

(D) they can join different DNA fragments.

Open solution

Q142. AIPMT 1994

Which of the following organelles is related with genetic engineering?

(A) Mitochondria (B) Plasmids (C) Golgi bodies (D) Lysosomes

Open solution

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MCQ (More than One Correct Answer) 1 questions

Q143. NEET 2023

Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence.

Open solution

A. Insertion of recombinant DNA into the host cell

B. Cutting of DNA at specific location by restriction enzyme

C. Isolation of desired DNA fragment

D. Amplification of gene of interest using PCR

Choose the correct answer from the options given below :

(A) C, A, B, D **(B)** C, B, D, A **(C)** B, D, A, C **(D)** B, C, D, A

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **B**	3. **B**	4. **D**	5. **A**
6. **D**	7. **D**	8. **B**	9. **C**	10. **A**
11. **D**	12. **B**	13. **B**	14. **C**	15. **C**
16. **A**	17. **A**	18. **A**	19. **D**	20. **A**
21. **B**	22. **C**	23. **C**	24. **C**	25. **B**
26. **A**	27. **D**	28. **A**	29. **A**	30. **A**
31. **A**	32. **B**	33. **C**	34. **C**	35. **B**
36. **A**	37. **C**	38. **B**	39. **C**	40. **A**
41. **C**	42. **B**	43. **C**	44. **B**	45. **A**
46. **D**	47. **C**	48. **C**	49. **D**	50. **B**
51. **B**	52. **D**	53. **D**	54. **C**	55. **D**
56. **B**	57. **A**	58. **D**	59. **A**	60. **B**
61. **C**	62. **B**	63. **B**	64. **C**	65. **B**
66. **B**	67. **A**	68. **D**	69. **C**	70. **B**
71. **D**	72. **D**	73. **D**	74. **C**	75. **D**
76. **A**	77. **C**	78. **C**	79. **D**	80. **C**
81. **B**	82. **A**	83. **A**	84. **A**	85. **B**
86. **C**	87. **A**	88. **B**	89. **D**	90. **A**
91. **D**	92. **D**	93. **C**	94. **B**	95. **D**
96. **B**	97. **A**	98. **A**	99. **B**	100. **B**
101. **B**	102. **A**	103. **A**	104. **A**	105. **D**
106. **D**	107. **D**	108. **D**	109. **A**	110. **B**
111. **A**	112. **D**	113. **B**	114. **C**	115. **B**
116. **C**	117. **A**	118. **D**	119. **B**	120. **C**
121. **B**	122. **A**	123. **A**	124. **B**	125. **D**
126. **B**	127. **A**	128. **D**	129. **C**	130. **A**
131. **A**	132. **C**	133. **C**	134. **B**	135. **B**
136. **D**	137. **C**	138. **D**	139. **D**	140. **B**
141. **A**	142. **B**			

MCQ (More than One Correct Answer)

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143. **B, D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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